



# **Autonomous Drone Tutorial**

## **Flapper Drones**

Matěj Karásek, PhD  
Founder & Director Flapper Drones

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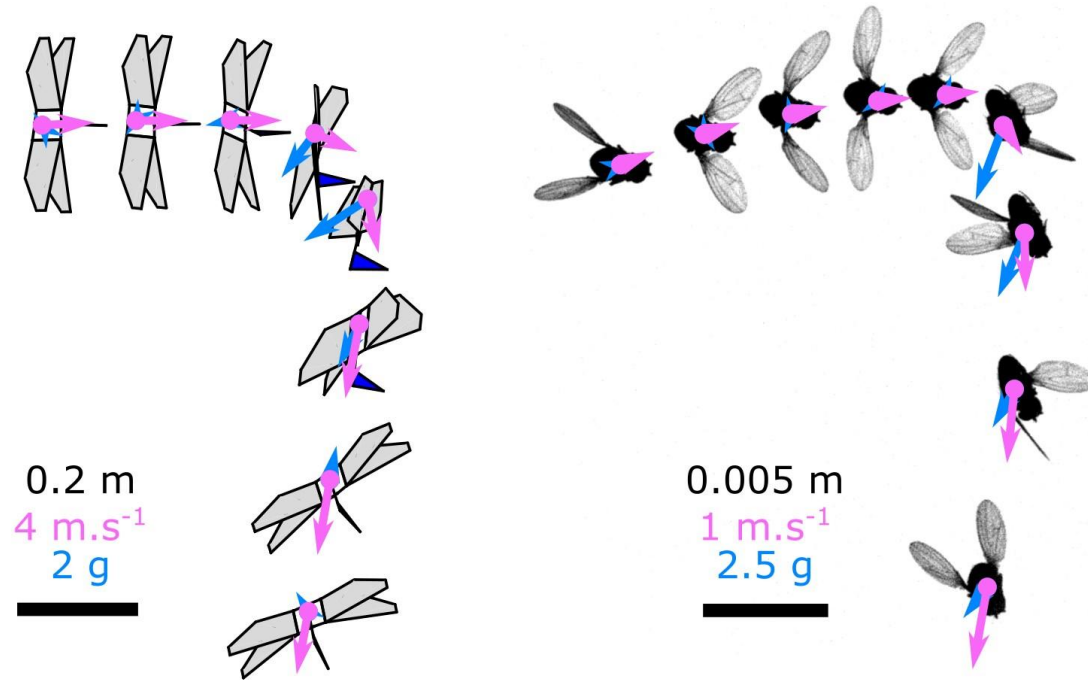
# About Flapper Drones

- Founded in 2019 in the Netherlands (Spinoff TU Delft)
- Since 2022 in Jablonec nad Nisou, Czechia
- Bioinspired R&D platforms
- Indoor drone performances (collaboration with Wombatronic)



# Research spinoff

## Evasive maneuvers of fruit flies



# Flapper Nimble+



## SPECIFICATIONS:

|                                   |                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------|
| <b>Min. takeoff weight:</b>       | 102 g                                                                                      |
| <b>Max. payload:</b>              | 30 g                                                                                       |
| <b>Wingspan:</b>                  | 49 cm                                                                                      |
| <b>Flapping frequency:</b>        | 13-15 Hz (hover)<br>20 Hz (max)                                                            |
| <b>Actuators:</b>                 | 2 x BL-DC<br>2 x rotary servo                                                              |
| <b>Battery:</b>                   | 300 mAh 2S LiPo                                                                            |
| <b>Flight time:</b>               | 8 minutes<br>5 minutes (hover with max. payload)<br>Up to 14 minutes with larger batteries |
| <b>Open-source firmware</b>       |                                                                                            |
| <b>Open-source Python API</b>     |                                                                                            |
| <b>Indoor Positioning Systems</b> |                                                                                            |





# Flight control hardware & optional sensors

## Crazyflie Bolt

STM32F4 MCU

- I2C, SPI, UART
- 4 x GPIO

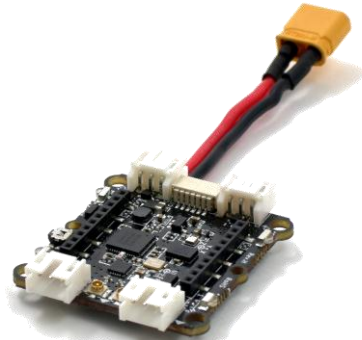
Bosch BMI088 IMU

nRF51822 SoC 2.4 GHz radio

External receiver

WS2812b RGB (3.8V, 1A)

microSD (data logging)



## Crazyflie sensor decks

Multi-laser Ranger (ToF)

Flow deck (optic flow)\*

AI deck\* (camera + processing)

UWB deck\*

*\* Experimental support*



## Indoor Positioning Systems

**HTC/Steam VR Basestations**

(via Crazyflie Lighthouse deck)

**Optical Motion Capture**

(OptiTrack, Vicon, Qualysis, ...)

**UWB Radio Beacons\***

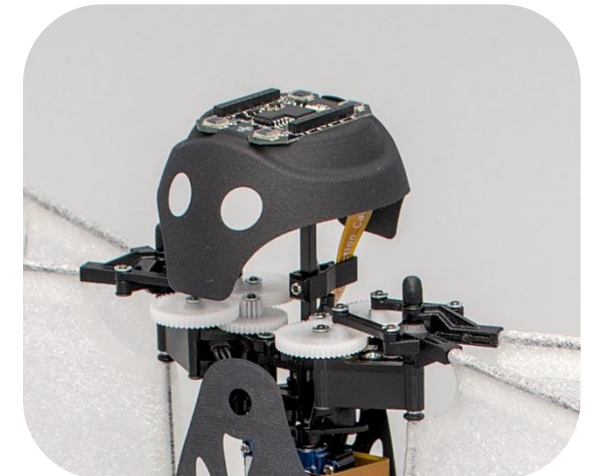
(via Crazyflie Loco Positioning System)



Directly mounted deck



Extension cables



Integration into body shells

# Tutorial setup

## Crazyflie Bolt

STM32F4 MCU

- I2C, SPI, UART
- 4 x GPIO

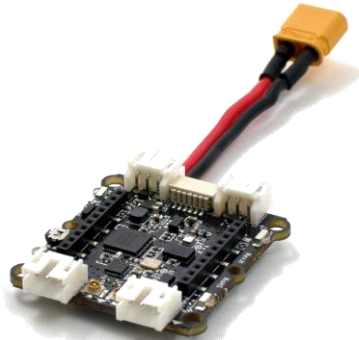
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External receiver

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## Crazyflie SW Ecosystem

Open-source firmware

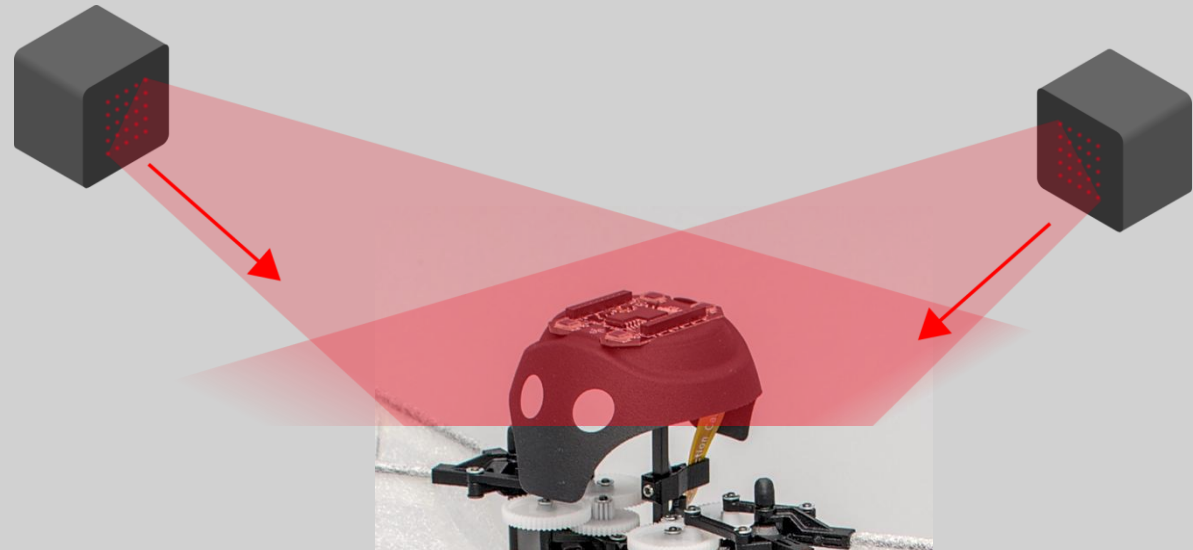
<https://github.com/bitcraze/crazyflie-firmware>

Open-source Python API

- High level control
- Swarming

<https://github.com/bitcraze/crazyflie-lib-python>

## *Lighthouse tracking system*



# Tutorial setup

- Lighthouse Basestations
  - **Do not bump into the tripods**, otherwise we will need to recalibrate...
- Geofence
  - The Flapper *should* not be able to fly out of a virtual cage
- LED status (top LEDs)
  - Color – battery level
  - Blinking – geofence activated (setpoint outside)
- Autoland
  - Flappers should autoland when low battery
- URI – *Unique Resource Identifier*
  - [DongleID]/[Radio Channel]/[Address]

# Connecting to Raspberry Pi

- 1) Download **TigerVNC** (or see shared folder for TigerVNC files)
- 2) Connect to „**Flapper WiFi**“  
Password: „NimblePlus“
- 3) Run Tiger VNC and connect to the following VNC server:  
**Group 1: „192.168.0.201“**  
**Group 2: „192.168.0.202“**
- 4) Confirm certificate exception (2x prompt)
- 5) Use these credentials:  
Username: „pi“  
Password: „password“





# Tutorial example scripts

- Example 0:
  - Python basics
- Example 1:
  - Connecting to the Flapper
  - Controlling flight using the position HL commander: left, right, up...
- Example 2:
  - Controlling flight using the High Level Commander: goto, yaw, spiral
  - Setting drone parameters (RGB Lights color)
- Example 3:
  - Logging data while flying and plotting it
  - Swarming class allowing to interact with multiple Flappers/Crazyflies
  - Using live sensor data of a hand-held Crazyflie (Magic Wand) to control a Flapper

All scripts available here:

<https://github.com/matejkarasek/FlapperTutorial>

# Example 1

- Run the script, link what you observe with the code
- Adjust the script to make the Flapper fly to your liking

## Example 2

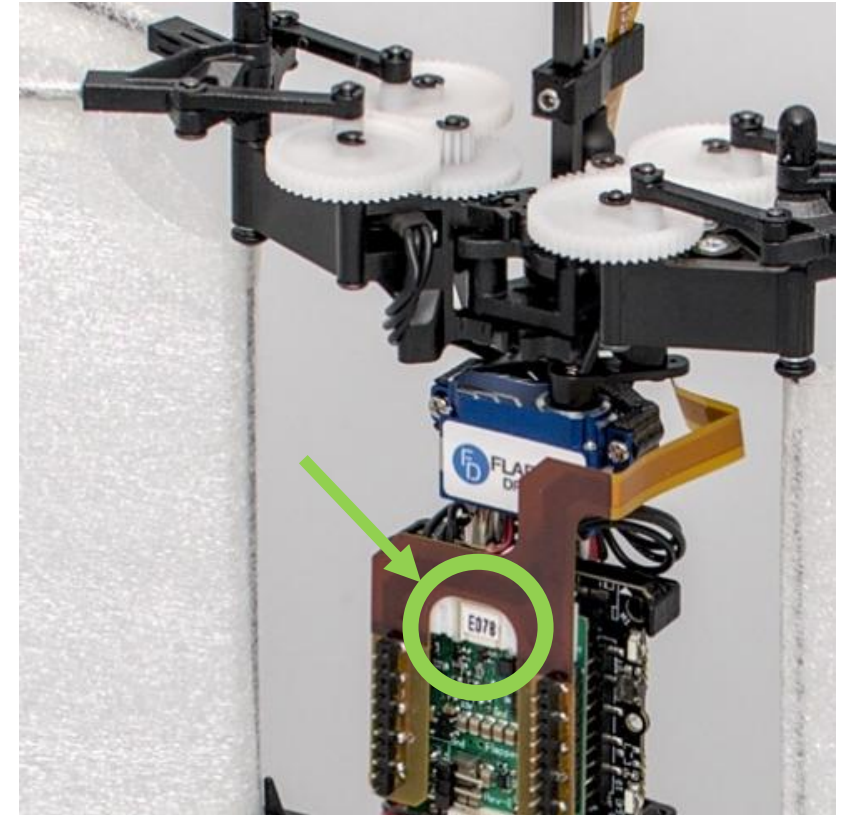
- Run the script, link what you observe with the code
- Adjust the script to make the Flapper fly to your liking
  - Did it work the way you wanted?
  - Iterate to get the desired flight trajectory
  - Add RGB light effects
  - Can you design a trajectory around a human standing in the center?

## Example 3

- Run the script, link what you observe with the code
- Adjust the script to make the Flapper respond to the „Magic Wand“ movements as you would like
  - Can you amplify the Magic Wand motion? (small wand movements → larger Flapper movements)
  - Can you also control the yaw angle?
  - Could the magic wand roll/pitch angles be used to steer the Flapper's velocity?
  - Could you use the onboard barometric pressure sensor to control height?

# URI

- URI – *Unique Resource Identifier*
  - radio://[DongleID]/[Radio Channel]/[Address]
  - Address: „FD##“  
where ## are the last 2 digits of the Flapper's ID
  - Group 1: Radio Channel 04
  - Group 2: Radio Channel 08





# Useful resources

- Crazyflie Python Library documentation:

<https://www.bitcraze.io/documentation/repository/crazyflie-lib-python/master/>

- List of logging variables:

<https://www.bitcraze.io/documentation/repository/crazyflie-firmware/master/api/logs/>

- List of parameters that can be set:

<https://www.bitcraze.io/documentation/repository/crazyflie-firmware/master/api/params/>

- Crazyflie Python Library source code:

<https://github.com/bitcraze/crazyflie-lib-python/tree/master/cflib>

- Flapper Nimble+ documentation

<https://docs.flapper-drones.eu/doku.php?id=nimbleplus:specs>